

ISutra — System Architecture & Technical Specification

AI Perception vs. Deterministic Decision Logic in High-Stakes Public Procurement

Smart India Hackathon (SIH) Problem Statement SIH26108

1. Executive Summary & Design Principles

In government procurement, decisions directly impact public safety, infrastructural integrity, and statutory accountability under the **General Financial Rules (GFR, 2017)** and the **Central Vigilance Commission (CVC)** guidelines. An algorithmic recommendation engine that invents standards or guesses safety clauses exposes procurement authorities to severe legal and financial risks.

ISutra solves this through a fundamental architectural separation:

$$\text{\textbf{AI Perception Layer}} \rightarrow \text{\textbf{Human-in-the-Loop Barrier}} \rightarrow \text{\textbf{Deterministic Decision Engine}} \rightarrow \text{\textbf{Official BIS Ground Truth}}$$

- AI Perception Layer:** Generative AI (Google Gemini 2.5 Flash, OpenAI GPT-4o-mini) and NLP parsers are used **only** to read unstructured text/PDFs and extract structured parameters.
- Strict Anti-Hallucination Guardrail:** AI models are **strictly forbidden** from generating standard numbers, fabricating safety clauses, inventing amendment dates, or issuing legal compliance certificates.
- Deterministic Decision Engine:** Recommendations are calculated using a transparent, reproducible, mathematical 6-factor linear model.
- End-to-End Auditability:** Every recommendation includes an unbroken 5-stage chain of custody linking the tender text directly to verified BIS gazette URLs.

2. Five-Layer System Architecture

```
graph TD
  subgraph Layer 1: Perception Layer
    A1[Tender Text / Clauses] --> B[Multilingual Normalization]
    A2[PDF / DOCX Ingestion] --> B
    B --> C[NLP / AI Extractor]
    C --> D[20+ Structured Dimensions]
  end

  subgraph Layer 2: Human-in-the-Loop Review Barrier
    D --> E[Requirement Review Interface]
    E -->|Officer inspects, edits & adds| F[Confirmed Requirements State]
  end

  subgraph Layer 3: Deterministic Decision Engine
    F --> G[6-Factor Scorer: standardsMatcher.ts]
    G --> H[Product / Category Match: 30%]
    G --> I[Keywords, Title & Scope: 25%]
    G --> J[Application Domain Fit: 15%]
    G --> K[Environmental Context: 10%]
    G --> L[Technical Parameters: 10%]
    G --> M[Safety & Testing: 10%]
    G --> N[Specificity Tiers 1-6 & Contradiction Filter]
  end

  subgraph Layer 4: Intelligence & Auditability Layer
    N --> O[Ranked BIS Candidates]
    O --> P[4-State Reference Coverage Gap Analyzer]
    O --> Q[Allied & Normative Standards Resolver]
    O --> R[5-Stage Lifecycle Stepper & Amendments]
    O --> S[Regulatory & QCO Assessment Layer]
    O --> T[11-Dimension Neutral Comparator]
    O --> U[Printable Procurement Report]
  end

  subgraph Layer 5: Statutory Ground Truth
    P & Q & R & S --> V[Official BIS Gazette & Portals: bis.gov.in, egazette.in]
  end
```

3. Layer-by-Layer Detailed Specification

Layer 1: Ingestion & Perception Layer

Handles diverse, unstructured inputs and normalizes them into structured engineering concepts.

- **Document Extraction Service** (`documentExtractionService.ts`):
 - Inspects file magic-bytes (`%PDF-` , `PK\x03\x04` for DOCX) to prevent malicious disguise attacks.
 - Extracts text streams preserving paragraph breaks and heading hierarchies.
 - Detects scanned image-only PDFs and safely rejects them with HTTP `422` and remediation instructions.

- Dynamic OCR hook architecture (`ENABLE_OPTIONAL_OCR`) with provenance tracking (`isOcrDerived: boolean`).
- **Multilingual Normalization Layer** (`multilingualService.ts`):
 - Performs Unicode script range detection for Devanagari (Hindi) and Telugu.
 - Maps Indic technical terms (e.g., "स्ट्रीट लाइट ल्यूमिनेयर", "స్ట్రీట్ లైట్ లూమినైర్") to canonical English taxonomy (`Luminaires` , `Roadway Lighting`) while preserving the original Indic source snippet.
 - Guarantees deterministic parity: An identical tender specification in English, Hindi, or Telugu produces the exact same standard recommendation score.
- **AI / NLP Parameter Extractor** (`nlpExtractor.ts` , `aiClient.ts`):
 - Dynamically routes requests across Google Gemini, OpenAI, or a built-in offline regex NLP engine.
 - Extracts 20+ structured procurement dimensions:
 - Core taxonomy: `product_name` , `category` , `subcategory` .
 - Operational context: `application_domain` , `operating_environment` , `installation_type` .
 - Quantitative specifications: `electrical_ratings` , `dimensional_limits` , `material_composition` .
 - Compliance parameters: `safety_features` , `testing_requirements` , `performance_metrics` .

Layer 2: Human-in-the-Loop Review Barrier

Public procurement law requires that tender drafting officers take responsibility for specifications.

- **Review UI** (`RequirementReviewPage.tsx`):
 - Displays the extracted parameter hierarchy with confidence indicators.
 - Allows officers to edit numerical values, delete misidentified terms, and add missing criteria.
 - Flags uncertain parameters with `needs_review` .
- **State Enforcement:** The backend controller (`analysisController.ts`) guarantees that the recommendation engine **only reads from the confirmed requirement state**.

Layer 3: Deterministic 6-Factor Decision Engine

The core scoring engine (`standardsMatcher.ts`) calculates a normalized relevance score ($S \in [0.0, 1.0]$) using a linear combination of six calibrated factors:

$$S = \sum_{i=1}^6 w_i \cdot s_i = 0.30 \cdot s_1 + 0.25 \cdot s_2 + 0.15 \cdot s_3 + 0.10 \cdot s_4 + 0.10 \cdot s_5 + 0.10 \cdot s_6$$

Factor Weights & Evaluation Criteria:

FACTOR WEIGHT DISTRIBUTION		
[Product / Category Match: 30%]	→	Taxonomy & product classification
[Keywords, Title & Scope: 25%]	→	Gazette title & published scope
[Application Domain Fit: 15%]	→	Installation & environment
[Environmental Context: 10%]	→	Ingress, thermal, weather
[Technical Parameters: 10%]	→	Ratings, wattage, materials
[Safety & Testing: 10%]	→	Type tests, protection circuits

- 1. Product / Category Match (\$w_1 = 0.30\$):**
 - \$s_1 = 1.0\$: Exact category match (e.g. `Luminaires` → `Luminaires`).
 - \$s_1 = 0.5\$: Subcategory / related family match.
 - \$s_1 = 0.0\$: No categorical alignment.
- 2. Keywords, Title & Scope (\$w_2 = 0.25\$):**
 - Jaccard similarity and semantic token matching between user terms and the standard's official Title and published BIS Scope statement.
- 3. Application Domain Fit (\$w_3 = 0.15\$):**
 - Compares deployment application (e.g. *Roadway/Street lighting* vs *Underground mining* vs *Floodlighting*).
- 4. Environmental Context (\$w_4 = 0.10\$):**
 - Evaluates ambient conditions (e.g., outdoor weather, high temperature, corrosive/saline, dust ingress).
- 5. Technical Parameters (\$w_5 = 0.10\$):**
 - Compares physical and electrical ratings (voltage, wattage, conductor cores, steel grade).
- 6. Safety & Testing Specifications (\$w_6 = 0.10\$):**
 - Validates safety features (surge protection, flame retardancy) and mandatory laboratory tests (type tests, routine tests).

Specificity Hierarchy (Tiers 1–6):

To prevent generic standards from outranking specific equipment standards:

- **Tier 1:** Exact Product-Type & Application Match (e.g., IS 10322 Part 5 Sec 3 for Street Lighting).
- **Tier 2:** Specific Subcategory Standard.
- **Tier 3:** Broad Equipment Family Standard (e.g., IS 10322 Part 1 General Luminaire Requirements).
- **Tier 4:** Allied Normative Reference.
- **Tier 5:** General Test Method Standard (e.g., IS 10810 series for Cable Testing).
- **Tier 6:** Cross-Domain Generic Standard (e.g., IS/IEC 60529 Ingress Protection).

Contradiction Handling & Product-Form Conflicts:

When user requirements specify an application (e.g., *Outdoor Highway Roadway*), standards explicitly restricted to incompatible forms (e.g., *Underground Coal Mining Luminaires*) receive a severe contradiction penalty: $s_{\text{application}} = 0.0, \quad \text{status} = \text{'contradiction'}$ This prevents misleading high scores caused by generic keyword matches.

Layer 4: Intelligence & Auditability Layer

1. The 5-Stage Audit Traceability Chain

Every recommendation includes an unbroken audit trail: $\text{User Input} \rightarrow \text{Extracted Requirement} \rightarrow \text{Scoring Signal} \rightarrow \text{Standard Clause} \rightarrow \text{Official BIS URL}$

2. Four-State Reference Coverage Taxonomy

Implemented in `requirementGapAnalyzer.ts` :

- `supported` : Confirmed present in verified clauses.
- `not_supported` : Contradicted by standard scope.
- `not_available` : Dimension is outside the standard's scope (e.g., safety standard does not govern photometric efficiency).
- `needs_verification` : Parameter requires manual inspection in the complete physical standard.

3. Allied Standards Explorer

Implemented in `standardsRelationshipService.ts` :

- Resolves normative references, test methods, safety codes, and installation guides.
- Distinguishes **Verified Relationships** (backed by explicit clause citations and official URLs) from **Unclassified References**.

4. Standards Lifecycle & Gazette Stepper

Implemented in `standardLifecycleService.ts` :

- 5-stage stepper: `STANDARD` $\rightarrow$ `REVISION` $\rightarrow$ `AMENDMENT` $\rightarrow$ `REAFFIRMATION` $\rightarrow$ `VERIFIED PORTAL` .
- Tracks periodic 5-year reaffirmations by BIS sectional committees.

5. Regulatory & Certification Engine

Implemented in `regulatoryService.ts` :

- Evaluates mandatory **Quality Control Orders (QCOs)** issued by Ministries under the BIS Act 2016.
- Returns strictly 3 states: `VERIFIED APPLICABILITY` , `VERIFICATION REQUIRED` , or `NO VERIFIED REGULATORY RECORD` .

6. Neutral Multi-Dimensional Standard Comparison

Implemented in `standardsComparator.ts` :

- Compares 2 or 3 standards across 11 objective dimensions without subjective rankings.
-

Layer 5: Statutory Ground Truth

- Connects directly to official portals:
 - BIS Standards Portal: `https://services.bis.gov.in`
 - BIS Know Your Standards:
`https://www.services.bis.gov.in/php/BIS_2.0/bisconnect/knowyourstandards/`
 - Gazette of India: `https://egazette.gov.in`
 - MeitY CRS: `https://crsbis.in`
 - Every recommended standard displays direct hyperlinks to official source records.
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4. Ingestion Schema & Verification Protocol

Implemented in `standardsIngestionService.ts` :

```
export interface StandardIngestionPayload {
  id: string;           // e.g. "is-10322-5-3"
  standard_number: string; // e.g. "IS 10322 (Part 5/Sec 3): 2012"
  title: string;        // Official BIS Title
  category: string;     // Canonical Category
  scope: string;        // Non-empty official scope statement
  source_url: string;   // Strict https://*.gov.in or https://*.crsbis.in URL
  last_verified: string; // ISO 8601 UTC timestamp
  status: 'active' | 'withdrawn' | 'superseded' | 'under_revision';
}
```

Records that fail government domain verification, contain invalid standard syntax, or lack verification dates are strictly rejected.

5. Sequence Diagram: Complete Recommendation Flow

```
sequenceDiagram
    autonumber
    actor Officer as Procurement Officer
    participant Web as Frontend (React 19)
    participant API as Backend (Express)
    participant DocSvc as Document Service
    participant AI as AI Perception Layer
    participant Matcher as Deterministic Matcher
    participant DB as MongoDB / Verified Dataset
    participant BIS as Official BIS Portals

    Officer->>Web: Uploads Tender PDF or pastes text
    Web->>API: POST /api/analysis/document or /api/analyze
    API->>DocSvc: Validate magic bytes & extract text
    DocSvc-->>API: Clean text stream
    API->>AI: Extract 20+ procurement dimensions
    AI-->>API: Extracted structured parameter tree
    API-->>Web: Return Analysis Session ID & parameters
    Web->>Officer: Display Human-in-the-Loop Review UI
    Officer->>Web: Edits, adds & confirms parameters
    Web->>API: PUT /api/analysis/:id/requirements (confirmed=true)
    API-->>Web: Confirmation ACK
    Web->>API: GET /api/analysis/:id/recommendations
    API->>DB: Fetch verified BIS catalog (40 records)
    API->>Matcher: Run 6-Signal Scorer & Specificity Tiers
    Matcher-->>API: Ranked results with mathematical breakdowns
    API-->>Web: Return recommendations & factor proofs
    Officer->>Web: Clicks "Why This Standard?" & "Gap Analysis"
    Web->>API: GET /api/analysis/:id/recommendations/:stdId/gap-analysis
    API-->>Web: 4-State Reference Coverage Breakdown
    Officer->>Web: Clicks official BIS verification link
    Web->>BIS: Direct navigation to official bis.gov.in gazette record
```